

Converse, Contrapositive, and Inverse

- The proposition $q \Rightarrow p$ is called the **converse** of $p \Rightarrow q$.
- The proposition $\neg q \Rightarrow \neg p$ is called the **contrapositive** of $p \Rightarrow q$.
- The proposition $\neg p \Rightarrow \neg q$ is called the **inverse** of $p \Rightarrow q$.

Logically Equivalent

Definition: When two compound propositions always have the same truth value we call them **equivalent**.

- The symbol $\equiv$ is used to denote the logical equivalences.
- $p \equiv q$ denotes the two compound propositions p and q are logically equivalent.
- Note that the symbol $\equiv$ is not a logical connective, and $p \equiv q$ is not a compound proposition.

Biconditional

Definition: The biconditional. Let p and q be statements. The biconditional,

$p \Leftrightarrow q$, read as “ p if and only if q ”, is true when p and q have the same truth value and false when p and q take different truth values.

The biconditional $p \Leftrightarrow q$ is also sometimes written as

- p iff q ,
- p is equivalent to q .
- p is a necessary and sufficient condition for q
- If p , then q , and conversely.

The Truth Table for the Biconditional ($p \Leftrightarrow q$)

p	q	$p \Leftrightarrow q$
T	T	T
F	T	F
T	F	F
F	F	T

Truth Tables of Compound Propositions

1. Create a table with 2^n rows if a compound proposition involves n propositions.
2. Fill in the first n columns with all possible combinations.
3. Determine and enter the truth value for each combination.

Precedence of Logical Operators.

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\Rightarrow$	4
$\Leftrightarrow$	5

Tautologies and Contradictions

- A compound proposition that is **always true**, no matter what the truth values of the propositions that occur in it, is called a **tautology**.
- A compound proposition that is **always false** is called a **contradiction**.
- A compound proposition that is neither a tautology nor a contradiction is called a **contingency**.

Some useful logical equivalences.

(i) $\sim(\sim p) \equiv p$

(ii) $\sim(p \wedge q) \equiv \sim p \vee \sim q$

(iii) $\sim(p \vee q) \equiv \sim p \wedge \sim q$

(iv) $p \Rightarrow q \equiv \sim p \vee q$

(v) $\sim(p \Rightarrow q) \equiv p \wedge \sim q$ ✗

Proof of (iv): Let $S: p \Rightarrow q$ and
 $T: \sim p \vee q$

Case (i) when p is T and q is F:

$p \Rightarrow q$ is F (since p is T and q is F)
 i.e. S is F
 $\sim p$ is F (since p is T)

$\therefore \sim p \vee q$ is F (since $\sim p$ is F and q is F)
 i.e. T is F

$\therefore$ the truth values of S and T are same.

Ex:

- Suppose the value of $p \Rightarrow q$ is T; what can be said about the value of $\sim p \wedge q \Leftrightarrow p \vee q$?
- Suppose the value of $p \Leftrightarrow q$ is T; what can be said about the values of $p \Leftrightarrow \sim q$ and $\sim p \Leftrightarrow q$?
 - Suppose the value of $p \Leftrightarrow q$ is F; what can be said about the values of $p \Leftrightarrow \sim q$ and $\sim p \Leftrightarrow q$?
- Construct the truth table for each of the following statements.
 - $p \Rightarrow (p \Rightarrow q)$.
 - $(p \vee q) \Leftrightarrow (q \vee p)$.
 - $p \Rightarrow \neg(q \wedge r)$.
 - $(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$.
 - $(p \Rightarrow (q \wedge r)) \vee (\neg p \wedge q)$.
 - $(p \wedge q) \Rightarrow ((q \wedge \neg q) \Rightarrow (r \wedge q))$.

The Algebra of Propositions

Propositions, under the relation of logical equivalence, satisfy various laws or identities, which are listed below.

1. Idempotent Laws

- a. $p \equiv p \vee p$.
- b. $p \equiv p \wedge p$.

2. Commutative Laws

- a. $p \wedge q \equiv q \wedge p$.
- b. $p \vee q \equiv q \vee p$.

3. Associative Laws

- a. $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.
- b. $p \vee (q \vee r) \equiv (p \vee q) \vee r$.

4. Distributive Laws

- a. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- b. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

5. De Morgan's Laws

- a. $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
- b. $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

6. Conditional identity

$$p \Rightarrow q \equiv \neg p \vee q$$

7. Complement Law

$$\neg(\neg p) \equiv p.$$

Ex prove that.

8. Identity laws

- a. $p \wedge \mathbf{T} \equiv p$
- b. $p \vee \mathbf{F} \equiv p$

(i) $\neg(p \Rightarrow q) \equiv p \wedge \neg q$
(ii) $p \Rightarrow q \equiv \neg q \Rightarrow \neg p$

9. Domination laws

- a. $p \vee \mathbf{T} \equiv \mathbf{T}$
- b. $p \wedge \mathbf{F} \equiv \mathbf{F}$

10. Negation laws

- a. $p \vee \neg p \equiv \mathbf{T}$
- b. $p \wedge \neg p \equiv \mathbf{F}$

Example:

Show that $p \wedge \sim(q \Rightarrow \sim r) \equiv p \wedge (q \wedge r)$

Ex. Using logically equivalent statements, without the direct use of truth tables,

1. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent
2. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.
3. Show that $(p \wedge q) \Rightarrow (p \vee q)$ is a tautology.
4. Show that $\sim(\sim p \wedge q) \wedge (p \vee q) \equiv p$

End of D₄